



Question 1

Max. score: 6.00

Consider you are working on Data Structures. You are given two strings s1 and s2. Your task is to convert the string s1 into s2 by inserting, removing, or replacing any character any number of times.

Example 1:

Input: s1 = "curs", s2 = "car"

Operations:

1. Remove 's' from s1
2. Replace 'u' with 'a'

Output: Minimum number of operations required to convert string s1 to s2 is 2

Example 2:

Input: s1 = "cars", s2 = "car"

We can convert s1 into s2 by removing 's'.

Output: Minimum number of operations required to convert string s1 to s2 is 1

So you implemented a function Minimum_Operations that takes the s1 and s2 as input and n,m where n is the length of s1, m is the length of s2, and returns the minimum number of operations required to convert String s1 into s2. And also the function Minimum_Operations takes a 2 Dimensional array named DP of size n*m with each value -1.

Pseudocode:

```
FUNCTION Minimum_Operations(s1, s2, n, m, dp)
    IF (n == 0) :
        RETURN m
    IF (m == 0) :
        RETURN n
    IF (dp[n][m] != -1) :
        RETURN dp[n][m];

    INT m1,m2,m3
```



Question 2

Max. score: 4.00

Alice, Bob, Ben and some of their friends are participated in a competition. They were standing in line to take the participation ticket. Each person were given with a unique ticket value which has a rating. You need to implement a code to form a team of 3 friends amongst them under the given rules:

1. Select 3 friends with index(a,b,c) with ticket value(ticket[a],ticket[b],ticket[c].
2. Team is valid only if(ticket[a] < ticket[b] < ticket [c] OR ticket[a] > ticket[b] > ticket [c] where,(0<=a<=b<=c<=n) (n=number of students participated)

Complete the below pseudocode:

Pseudocode

```
Function numTeams
{
  Input int ticket, int Result = 0;
  for(int mid = 1; mid< ticket.length -1;mid++)
  {
    int left=0;int right =0;
    int upperleft=0;int upperright =0;
    for(int b=0;b<ticket.length;b++)
    {
      if(b<mid and ticket[b]< ticket[mid]){
        left+=1;
      }
      else if(b>mid and ticket[b] > ticket[mid])
      {
        right+=1;
      }
    }
    //code line 18
  }
```



Question 3

Max. score: 2.00

Ben is working on Data Structures. He implemented a program in which there is a *bi-directional* graph with n vertices. Every vertex pair is connected by *at most one* edge, and no vertex has an edge to itself. He wants to determine if there is a *valid path* that exists from vertex `source` to vertex `destination`. He wants to return `true` if there is a *valid path* from `source` to `destination`, or `false` otherwise. Help him to complete the below pseudocode:

Pseudocode

```
Function isPossible():
    declare source;
    declare destination;
    declare adj[][]; // adj[i] contains source adjacent to source i
    declare stack[] = [source];
    declare visited[];
    while stack is not empty:
        source= stack.pop()
        if source == destination:
            return True
        if source in visited:
            continue
        add source to visited
        //Line12
        //Line13
    return False
```

What will be the code in place of lines 12 and 13?

Options

1. for state2 in adj[state-1]: //line 12
 stack.append(state) //line 13

What will be the code in place of lines 12 and 13?

Options

1. `for state2 in adj[state-1]:` `//line 12`
 `stack.append(state)` `//line 13`

2. `for state2 in adj[state]:` `//line 12`
 `stack.append(state)` `//line 13`

3. `for state in adj[state]:` `//line 12`
 `stack.append(state)` `//line 13`

4. `for state2 in adj[state]:` `//line 12`
 `stack.append(state2)` `//line 13`

☐ 1

☐ 2

☐ 3

☒ 4

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Question 7

Max. score: 6.00

Bob loves to play with numbers but his favorite one is 0. He was given an array of numbers and was asked to print the number of subarrays possible which sum up to 0.

Example

Input: arr = [6, 3, -1, -3, 4, -2, 2, 4, 6, -12, -7]

Output: 5

```
int hacker(int arr[], int n)
{
    unordered_map<int, list<int> > map
    int res=0
    int s= 0
    for (int i = 0; i < n; i++){
        s += arr[i]
        if (s == 0)
            res++;
        if (map.find(s) != map.end())
            list<int> l= map[s]
            //Line 6
        map[s].push_back(i)
    }
    return res
}
```

He came up with the code, but some code in between went missing. Can you help him replace *//Line 6* from the given options?

Options

1.

```
for (auto it = l.begin(); it != l.end(); it++)
    res++;
```



1

```
for (auto it = l.begin(); it != l.end(); it++)  
    res++;
```

2

2.

3

```
for (auto it = l.begin(); it != l.end()-1; it++)  
    res++;
```

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3.

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```
for (auto it = l.begin(); it != l.end(); it++)  
    res+=2;
```

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4.

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```
for (auto it = l.begin(); it != l.end(); it++)  
    res--;
```

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☐ 1

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☐ 2

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☐ 3

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☐ 4

Question 8

Max. score: 2.00

Jones and Anitha are playing with Matrices. Their teacher asked them to tell whether multiplying those two matrices is possible or not? Help Jones and Anitha to solve this problem. Complete the below pseudocode

Pseudocode:

```
//A is no of rows in matrix1
//B is no of rows in matrix2
//C is no of columns in matrix1
//D is no of columns in matrix2
function Solve(A,B,C,D)
    //condition line 6
    write "Possible"
    else
    write "Not possible "
```

What will be the code in place of line 6?

Options:

1. check if A==B then
2. check if D==B then
3. check if A==C then
4. check if C==B then

Analyse the below pseudocode and choose the option that represents the correct output and the Time Complexity of the Pseudocode?

Pseudocode

```
INTEGER arr[], counter
SET counter = 0
SET arr[]={343, 243, 24396, 57, 51246, 648}
INTEGER size = 6

FOR i -> 0 to size
    IF arr[i]%3 == 0 && arr[i]%9 == 0
        counter ++
    ENDIF
ENDFOR

PRINT counter
PRINT counter % 3
PRINT counter % 2
```

☐ Output : 5, 2, 3 , Complexity : $O(\text{size})$

☐ Output : 3, 0, 2 , Complexity : $O(1)$

☐ Output : 3, 0, 1 , Complexity : $O(1)$

In Data Structures you are working on **Single Linked List**. The below pseudocode contains a function **size_of** that takes the **head node(h)** of a single linked list as **input** and returns the size of the single linked list as **output**.

Pseudocode

```
class Node {  
    // constructor...  
    Node(value) {  
        data = value  
        next = NULL  
    }  
}  
  
FUNCTION size_of( h )  
    IF h==NULL  
        RETURN 0  
    ENDIF  
    // Line 11  
ENDFUN
```

Note : In the above pseudocode the **class Node** represents the structure of each **node** in the single linked list.

Now, which of the following code snippets best fit in the above pseudocode in **Line 11** such that the function **size_of** returns the correct size of the single linked list as output?



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2.

```
RETURN 1+size_of( h )
```

3.

```
RETURN 1+size_of( h.next )
```

4.

```
RETURN size_of( h )
```

☐ 1☐ 2☒ 3☐ 4[↶ Reset Answer](#)

There are n seas. Some of them are connected, while some are not. If sea A is connected directly with sea B, and sea B is connected directly with sea C, then sea A is connected indirectly with sea C. An *ocean* is a group of directly or indirectly connected sea and no other sea outside of the group. Return the total number of **oceans**. Complete below pseudocode:

Pseudocode

```
Function traversal() {
    for j from 0 to n {
        if (Mat[i][j] == 1 && visited[j] == 0) then
            visited[j] = 1;
            //Line 6
        }
    }
}

Function findOceans() {
    declare Mat[][], visited[];
    declare count = 0;
    declare n; // number of nodes
    for i from 0 to n {
        if (visited[i] == 0) {
            traversal(Mat, visited, i);
            count=count+1;
        }
    }
    return count;
}
```

What will be the code in place of line 6?



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```
        count=count+1;
    }
}
return count;
}
```

What will be the code in place of line 6?

Options:

1. traversal(Mat, visited, i);
2. traversal(Mat, visited, j);
3. traversal(Mat, visited, i+1);
4. traversal(Mat, visited, j+1);

☐ 1

☐ 2

☐ 3

☐ 4

[↶ Reset Answer](#)

Jones is learning Data Structures. His professor asked him to work on the concept of queues. He started to design a Circular queue that performs all operations. But he missed some lines of code. Help him to complete the below pseudocode:

Pseudocode

```
Function Node {  
    Input int value;  
    Input nextNode;  
  
    Function Node(int value) {  
        nextNode = null;  
    }  
}  
Function MyCircularQueue {  
  
    Input head, tail;  
    Input int count;  
    Input int capacity;  
  
    Function MyCircularQueue(int k) {  
        capacity = k;  
    }  
  
    Function boolean enqueue(int value) {  
        if (count == capacity)  
            return false;  
  
        newNode = value;
```



Question 15

Max. score: 50.00

Minimum execution time

You are given 3 arrays where each denotes some programs to be executed. The value of an array denotes the time required to execute the program.

You do the following operation until all three arrays become empty:

1. Take a program from the beginning of some non-empty array
2. Execute this program and remove this from that array

Assume, that the final order of executing comes out to be $[p_1, p_2, p_3, p_4, p_5, \dots]$, where p_i is the execution time of the i^{th} program in the final order.

Now, you define the waiting time of the i^{th} program in the final order as the summation of the execution times of the program which were executed before it that is, $w_i = p_1 + p_2 + \dots + p_{i-1}$

The final cost of all programs execution in the order equals $[w_1 + w_2 + w_3 + \dots]$.

Task

Determine the minimum cost after executing all programs.

Example

Assumptions

- $k1 = 2$
- $A_1 = [2, 1]$
- $k2 = 1$
- ...

New Submission All Submissions

Auto-complete connection failed. [Reconnect](#)

Save

C++17 (g++ 10.3.0)



```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 long long solve (int k1, vector<int> A1, int k2, vector<int> A2, int k3, vector<int> A3) {
5     // code starts here
6
7     int dp[k1+1][k2+1][k3+1];
8     long long ans=LL_MAX;
9     int n1=A1.size();
10    int n2=A2.size();
11    int n3=A3.size();
12    long long sum=0;
13    for(int i=0;i<n1;i++){
14        for(int j=0;j<n2;j++){
15            for(int k=0;k<n3;k++){
16
17            }
18        }
19    }
20
21 }
22
23 int main() {
24
25     ios::sync_with_stdio(0);
26     cin.tie(0);
27     int k1;
28     cin >> k1;
```

16:13 vscode



Test against custom input ▼

Custom Input populated

Compile & Test code

Submit code

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Question 14

Max. score: 50.00

Maximize friendship power

There are N cities and M bi-directional roads connecting some/all of them. 2 cities are connected if they are reachable from each other using the roads. Connected cities form a *kingdom*. Therefore, you can have at least 1 kingdom (if all are connected) or N kingdoms ($M = 0$).

There are N groups of people. The i^{th} group has A_i people. You take a *permutation* array B of $[1..N]$ and assign the i^{th} group to the $B[i]^{\text{th}}$ city. All people living in a kingdom are friends. The friendship power of a kingdom is defined as the number of different friendship pairs in the kingdom.

Task

Choose a permutation array B to maximize the sum of friendship power over all the kingdoms. The maximum friendship value can be very large. Therefore, return the value modulo 998244353.

Notes

- Assume 1-based indexing.
- Do not count the same friendship pairs twice. That is if 'x' and 'y' are friends, count this pair as one.
- 2 cities are connected if they are reachable from each other using the roads.
- The sequence of N integers is called a *permutation* if it contains all integers from 1 to N exactly once.
- Roads are given in form of 2D array *roads* whose element $[u, v]$ denotes that there is a bi-directional edge between u and v .
- x modulo y means the remainder when x is divided by y . In particular, if x modulo $y = z$,

New Submission

All Submissions

Auto-complete ready!

Save

C (gcc 10.3)

```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int maxFriendshipPower (int N, int M, int** roads, int* A) {
6     // Write your code here
7 }
8
9
10 int main() {
11     int T;
12     scanf("%d", &T);
13     for(int t_i = 0; t_i < T; t_i++)
14     {
15         int N;
16         scanf("%d", &N);
17         int M;
18         scanf("%d", &M);
19         int i_roads, j_roads;
20         int **roads = (int **)malloc((M)*sizeof(int *));
21         for(i_roads = 0; i_roads < M; i_roads++)
22         {
23             roads[i_roads] = (int *)malloc((2)*sizeof(int));
24         }
25         for(i_roads = 0; i_roads < M; i_roads++)
26         {
27             for(j_roads = 0; j_roads < 2; j_roads++)
28             {
```



Test against custom input ▼

Custom input populated ↺

Cor

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